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Evaluation of Segment Anything Model for Riparian Land Cover Classification from Aerial Imagery

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Abstract. Riparian land cover classification (LCC) is an important task for environmental monitoring and management, as it provides information on the state and characteristics of riparian zones, which are the transitional areas between terrestrial and aquatic ecosystems. However, LCC is challenging due to the high diversity and complexity of riparian vegetation and land use. In this paper, we evaluate the performance of Segment Anything Model (SAM), a recently proposed deep learning model for general image segmentation, for riparian LCC from aerial imagery. SAM can produce high quality object masks from input prompts such as points or boxes, and can generate masks for all objects in an image without fine-tuning. The authors compared SAM with the results derived from the DeepLabV3+ model trained by the dataset collected in Japan. Our experiments showed that SAM achieved competitive or superior results compared to other models, and can segment various riparian land cover types with high accuracy and consistency. The authors concluded that SAM is a promising tool for riparian LCC from aerial imagery, and suggest some directions for future research combining with the other model.

Keywords: DeepLabV3+, Land Cover Classification, Segment Anything Model, UAV

1 Introduction

Recent years, imagery derived from unmanned aerial vehicle (UAV) become an important tool for environmental monitoring. LCC mapping based on the UAV imagery is an efficient and critical approach for environmental researchers to collect the information on the specific location, especially the riparian area that are difficult for on-site survey. In this context, the authors evaluated SAM (segment anything model, a state-of-the-art segment model)-assisted results with DeepLabV3+ model [1] on classifying the land cover of the same locations.

2 Work flow

The authors trained the DeepLabV3+ model using land cover dataset (train/valid) that include cropped ortho-photo and responding masks, and inferred the ortho-photo from 3 selected locations (with TL) using trained DeepLabV3+ model. SAM segmented ortho-photo at the same mentioned locations, and derived from the segmented mapping the authors re-classified the land cover, namely SAM-assisted LCC Results. As the final step shown in the Figure 1, comparing the LCC Results derived from DeepLabV3+ and SAM-assisted with TL, respectively.

3 Conclusion

As depicted in the Figure 2, the findings of this research reveal that the performance shown in CM (i.e., Low-Vegetation/Ground, High-Vegetation and Water) of SAM-assisted Results is better than DeepLabV3+ LCC Results. The collaboration of these mentioned approach is in the consideration for the future research.

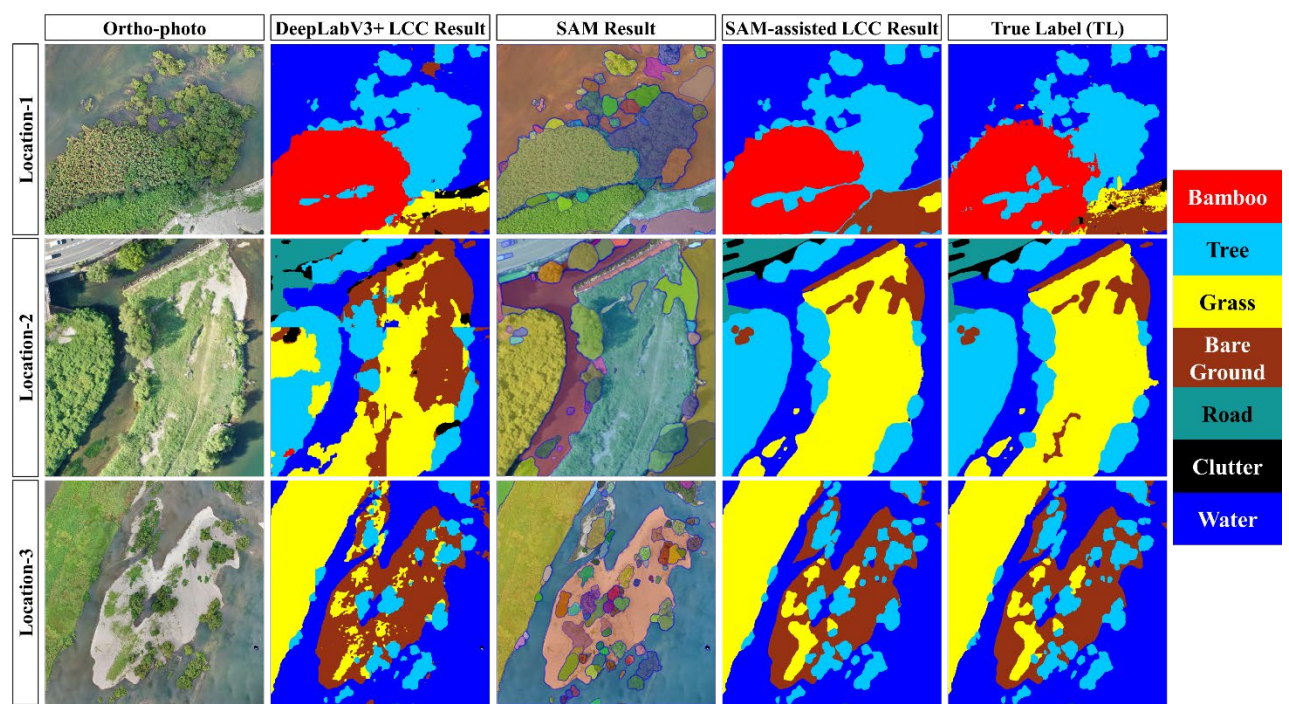


Figure 1 DeepLabV3+ and SAM-assisted LCC Results-based Comparison with Ortho-photo and TL of 3 locations.

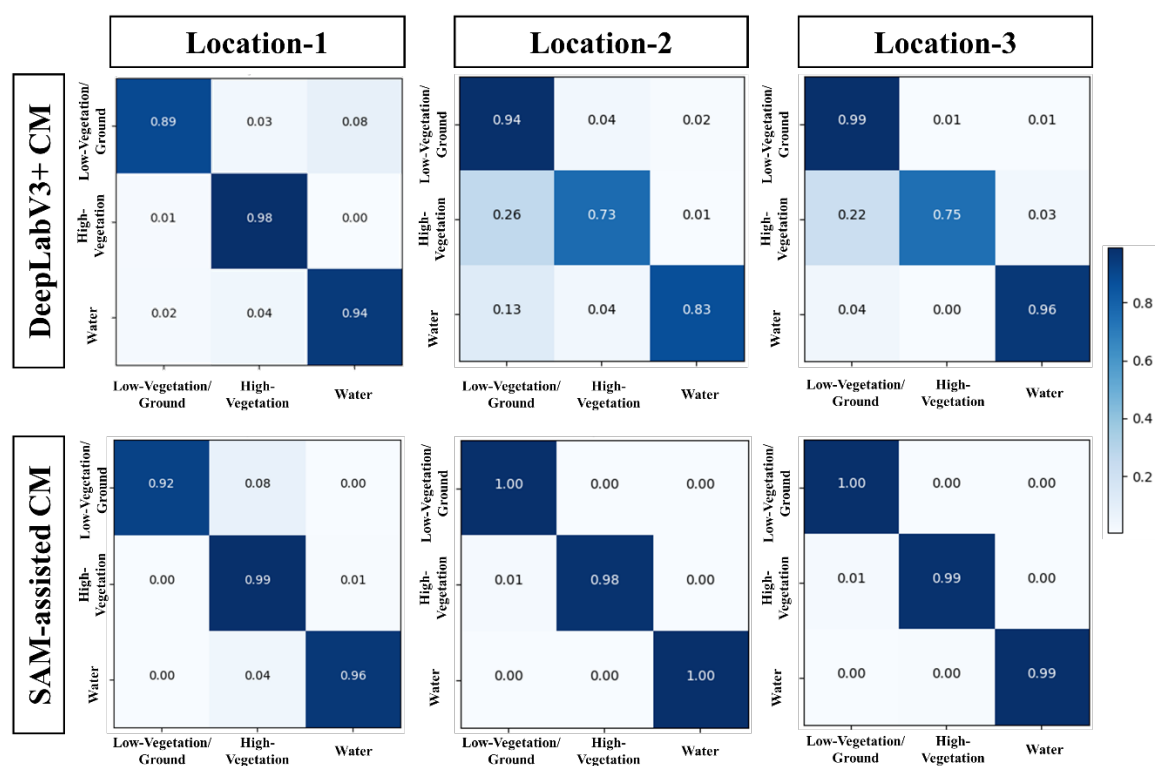


Figure 2 DeepLabV3+ and SAM-assisted LCC Confusion Matrix (CM) of 3 locations.

Reference

[1] Yoshida, Keisuke, et al. "Airborne LiDAR-assisted deep learning methodology for riparian land cover classification using aerial photographs and its application for flood modelling." *Journal of Hydroinformatics* 24.1 (2022): 179-201. 10.2166/hydro.2022.134